I. Experimental Evaluation

A. Experimental Setup

1) Data and Evaluation Protocol: We evaluate closed-set cross-receiver radio-frequency fingerprinting on the ManySig and ManyRx subsets of WiSig [1]. Source and target share the registered transmitter (TX) identities and have disjoint physical receiver (RX) sets. ManySig contains six TX, 12 RX and four acquisition dates: March 1, 8, 15 and 23, 2021 (D0–D3). We use equalized IQ, a central 256-sample window and float32 per-packet unit-RMS normalization before channel transformation, yielding a 2×256 input. Derived views of one packet retain its physical identity.

Table I lists the principal partitions. F0 has five source RX and seven target RX; its 15 source RX–date cells are not 15 physical receivers. F1/F2 change receiver membership while retaining the five/seven counts. For ManySig, packets ordered by stored physical indices within each source TX/RX/day cell are divided into 70 labeled training samples, 630 unlabeled samples and 300 validation samples. Thus L is 10% of the training pool, while the additional source-validation labels are disclosed separately. U labels remain hidden. Validation updates no model or persistent state.

The default ManySig source roles contain 6300 L, 56700 U and 27000 V packets; the one/five-percent training-label conditions reduce L while keeping its original physical source pool and V role. New F1/F2 replace four of five source receivers relative to F0 and have pairwise source-set intersection one. Every new F/G ancestor matches the new contract; historical F1/F2 receiver memberships are not relabeled. ManyRx retains missing and unequal cells, with 6533 L, 58828 U and 28023 V packets. Its source roles use per-cell floor fractions 0.07/0.63 and the remainder. The source-count study uses natural budgets and m=1,2,3,4,6; it does not fix L at 1080. ManySig and ManyRx share the WiSig acquisition system and are not independent capture campaigns.

Every test uses the same fixed received-IQ observations: clean plus high/mid/low suburban and high/mid/low urban synthetic channel conditions. These are simulated stress tests, not measured satellite-link validation. DURA is applied to every declared observation, including clean, without scene-aware bypass or additional output renormalization. Each model freezes every view and task before independent truth-side scoring. Target observations, labels and scores never enter training, hyperparameter selection, checkpoint selection or rerun decisions.

2) Complete Baselines and PAMF Training: We compare the complete methods RIEI [2], DRIFT-v2 [3], IB-RFF [4] and SDG [5]. Each retains its native objectives, jointly trained identity head and method-specific stopping rule (Table II). RIEI and DRIFT use the registered E100 implementation budgets; RIEI uses the registered short one-dimensional input stem. The comparison therefore evaluates complete recipes with their disclosed implementation budgets.

The four external Full methods use a preregistered 2-by-2 channel/ordinary-PL matrix. A/B disable the additional channel and C/D enable it; A/C disable PL and B/D enable it. The common auxiliary TX-CE slot has weight 0.68 in every cell; channel-off or unselected samples reuse the original L IQ, preserving its denominator and exposure. Native internal augmentation and Full losses remain active, including SDG two-view training. The channel-enabled input uses the actual practical ResidualChannel engine in residual/post-sync mode without equalization or output renormalization. Training draws only low suburban, mid urban and low urban with conditional probabilities 0.40/0.35/0.25. The application probability is 0.30/0.60/0.80 over maximum-budget intervals $[0, .20)$, $[.20, .45)$ and $[.45, 1)$, resetting within each native IB stage. Source-only early stopping can change realized exposure. PAMF-Sup has corresponding channel-off/on recipes with no U/EMA; native T1 retains its fixed original internal views and self-training. A deterministic FP32 input cache is shared by physical ID, epoch, stage and channel draw; no student targets or gradients are cached.

External semi-supervised variants use current-model hard pseudo-labels at confidence 0.95, PL weight 1 and U/L batch ratio 1, without EMA, neighbor gates or strong-view consistency. RIEI runs with zero pseudo-label acceptance at the fixed threshold remain in the reported matrix; U exposure and realized acceptance are retained in the training records. PAMF-T1 retains the fixed native schedule: ordinary/raw stochastic consistency, shallow/intermediate cross-RX CRIC, As cross entropy with As KL disabled, label smoothing 0.01, and native selective U/EMA training. Student and own EMA states are inherited, while optimizers and segment schedules reset. A logical PAMF epoch contains 222 updates and need not equal one pass over a changed source pool.

3) Restoration Permissions and Reporting: F denotes the identifier and G the DURA restorer. Supervised system comparisons use G-L: every G ancestor trains using labeled-source IQ only, without consuming its TX labels. Semi-supervised system comparisons use G-LU, which may also consume TX-hidden source U. G training uses simulated source-IQ pairs, without F weights/outputs or target data. All ancestors match the classifier's source contract; a restorer from a different RX/date/TX split is not reused as a matching teacher. Restorer cost includes its source-only training ancestry and is shared across frozen F consumers.

We report Accuracy and Macro-F1 over the full registered class set (six classes for ManySig and ten for ManyRx). S6 averages the six channel conditions within each seed before computing mean and sample SD across F seeds 2026092701/02/03. Three F seeds usually share one fixed G per contract: the reported SD is conditional F variation, not three independent complete-system replicas or a significance interval. A separate preregistered three-source F/G crossed-seed study checks joint-seed sensitivity. The evidence archive includes per-RX/TX/day results and confusion-derived repair/damage counts.

B. Complete Methods under Practical Residual Channels

All 627 registered readouts are complete: 528 newly trained external Full models, 66 PAMF supervised models, six new native-T1 models and 27 exact-contract native-T1 reuses. Each freezes two routes, three disjoint generalization populations

TABLE I
PRINCIPAL CONTRACTS WITH UNCHANGED D1–D3 TRAINING. THE THREE DISJOINT TEST POPULATIONS ARE SOURCE RX ON D0 (DATE), TARGET RX ON D1–D3 (RX), AND TARGET RX ON D0 (BOTH). D0 IS AN EARLIER UNSEEN DATE.

Contract	Source RX indices	TX	Date packets	RX packets	Both packets
F0	1,3,4,6,8	6	30000	126000	42000
F1 (new)	0,1,2,5,7	6	30000	126000	42000
F2 (new)	0,3,9,10,11	6	30000	126000	42000
ManyRx	Even indices 0–30	10	31279	92597	30424

TABLE II
REGISTERED TRAINING BUDGETS AND MODEL SELECTION. EXTERNAL SEMI-SUPERVISED VARIANTS ADD ORDINARY PL TO THE SAME COMPLETE METHOD. PAMF-T1 USES ITS NATIVE SELECTIVE SELF-TRAINING. NO POST-HOC CLASSIFIER FITTING IS PERFORMED.

Method	Budget and optimizer	Freeze and recipe
RIEI	E100; batch 64; SGD 10^{-4}	Final epoch; three optimizer steps per batch for the complete objective.
DRIFT-v2	E100; batch 256; Adam 10^{-4}	Final epoch; native adversarial, center and separation objectives.
IB-RFF	Two stages, each at most E200; batch 128; Adam 10^{-3}	Source-V minimum NLL; LR reduction patience 5; early-stop patience 10.
SDG	E200; batch 32; Adam 10^{-3}	Source-V minimum NLL; native two-view and contrastive objectives.
PAMF-Sup	E50/11100 updates; L batch 128; AdamW and segment cosine schedules	Fixed final checkpoint; supervised representation objectives; no U or EMA.
PAMF-T1	E50/11100 updates; L batch 128, U batch 256; AdamW and segment cosine schedules	13 labeled-only plus 37 joint epochs; own student/EMA inheritance across 20+10+20 stages.

and seven observations before independent scoring. The 600 new F models and four new eight-stage G lineages require no further training. We verify 742938 serialized confusion matrices and 371469 net-correction cells, reusing the previously verified F1/F2 overall cells. ManyRx uses ten classes and the other conditions six.

Table III retains the complete channel/ordinary-PL four-cell comparison on the F0 cross-RX population. Channel augmentation is not uniformly beneficial, and ordinary PL does not consistently improve a complete baseline. Raw-F and restored performance are therefore reported together. G-L is used by A/C and supervised PAMF, whereas G-LU is used by B/D and native T1; restored PL-group differences include this additional G-data permission. PAMF-T1 retains its own selective self-training and original internal views, so comparison against ordinary-PL baselines concerns complete recipes, not one isolated SSL loss.

1) Receiver and Date Generalization: The new F1 and F2 source sets each replace four of the five F0 source receivers, and their pairwise source-set overlap is one. Table IV separates earlier unseen-date evaluation on source RX, unseen RX on training dates, and their conjunction. Dates and receiver populations are not pooled. Native semi-supervision does not outperform the supervised recipe in every cell; all negative differences remain in the archive.

2) Label Availability, ManyRx and Source Count: Table V reports the one/five-percent training-label conditions, the default ten-percent condition and ManyRx. External rows use fixed D rather than selecting a cell from target scores. PAMF-Sup C and native T1 remain separate complete recipes. The source-count curves in Fig. 2 use m=1,2,3,4,6 with natural source budgets and the same six target RX: 108000 seen-date packets and 36000 D0 packets. No fixed-L budget or m=5 point is added; the seven-target F0 setting is not inserted into these curves.

The 600 new F models sum to 246.37 shared-worker elapsed hours; 27 reused F sources contribute 46.26 historical hours. Four new complete G lineages sum to 18.42 hours; the 17 other fully timed G lineages sum to 89.26 historical hours. Original fixed-F0 G-LU ancestor timings remain incomplete. The 3811 unique CPU augmentation-cache productions sum to 1.52 elapsed hours, the 22 restored-IQ caches to 1.17 hours and cached F predictions to 21.67 shared-worker hours. These are not exclusive GPU hours, CPU core-hours, online G+F latency or an end-to-end speedup. We retain all external augmentation/PL counts, including 61 zero-acceptance training rows. Complete native-PAMF gate acceptance and per-F realized channel counts are not recoverable from the compact telemetry and are left missing.

The following PAMF mechanism studies retain their original F0 all-target-date population (168000 packets/view). They are independently frozen historical mechanism evidence, not relabeled as the 126000-packet cross-RX endpoint above. Existing mechanism and timing measurements are reused without retraining or retesting.

C. PAMF Mechanism Analysis

1) Three Core Training Mechanisms: We group the identification objectives by function rather than observed gain. A is CRIC: shallow/intermediate same-TX, cross-physical-RX contrast. B is stochastic prediction consistency: ordinary128, ordinary64 and raw128 symmetric-KL terms switch together. C is native selective self-training: own EMA, confidence and adjacent/strong-view gates, pseudo-label CE and U entropy switch together. Table VI reports the complete model and one

TABLE III
F0 CROSS-RX TEST (D1–D3; 126000 PACKETS/VIEW). ACCURACY AND MACRO-F1 IN PERCENT, MEAN±SAMPLE SD OVER THREE F SEEDS. A/B/C/D DENOTE CHANNEL OFF/ON CROSSED WITH PL OFF/ON. NO CE-ONLY BASELINE IS INCLUDED.

Method	Cell	Raw clean	Raw S6	Restored clean	Restored S6	Restored S6 F1
RIEI	A	66.93 ± 0.71	52.43 ± 0.98	66.91 ± 0.69	62.43 ± 1.25	58.09 ± 1.49
DRIFT-v2	A	66.21 ± 3.30	25.54 ± 1.08	66.18 ± 3.25	60.25 ± 2.60	60.75 ± 2.75
IB-RFF	A	66.29 ± 1.40	14.21 ± 3.49	66.29 ± 1.38	62.20 ± 0.64	61.89 ± 0.74
SDG	A	74.25 ± 0.88	56.08 ± 3.95	74.29 ± 0.88	66.81 ± 0.74	66.77 ± 0.69
RIEI	B	67.28 ± 0.88	52.70 ± 0.84	67.33 ± 0.93	63.01 ± 1.44	58.77 ± 1.88
DRIFT-v2	B	67.69 ± 1.71	20.84 ± 1.57	67.64 ± 1.69	62.55 ± 2.23	62.97 ± 2.79
IB-RFF	B	66.44 ± 2.90	16.02 ± 2.91	66.47 ± 2.91	61.43 ± 1.79	61.00 ± 2.01
SDG	B	74.64 ± 0.14	57.01 ± 0.78	74.65 ± 0.13	68.48 ± 0.61	68.07 ± 0.46
RIEI	C	62.22 ± 1.53	50.08 ± 1.60	62.16 ± 1.49	57.59 ± 0.75	53.10 ± 0.85
DRIFT-v2	C	61.77 ± 6.73	55.22 ± 5.28	61.71 ± 6.71	57.05 ± 4.48	56.10 ± 5.14
IB-RFF	C	66.74 ± 2.61	53.10 ± 1.83	66.77 ± 2.59	62.72 ± 1.74	62.37 ± 1.80
SDG	C	73.18 ± 2.16	65.16 ± 2.03	73.25 ± 2.16	66.46 ± 1.33	66.56 ± 1.33
RIEI	D	61.63 ± 1.89	49.95 ± 1.55	61.68 ± 1.85	56.73 ± 0.91	52.34 ± 1.27
DRIFT-v2	D	66.12 ± 3.20	56.76 ± 2.29	66.11 ± 3.18	60.75 ± 4.42	58.46 ± 2.87
IB-RFF	D	66.63 ± 2.29	52.57 ± 2.17	66.65 ± 2.24	62.21 ± 2.86	61.74 ± 3.12
SDG	D	74.32 ± 1.52	65.30 ± 1.14	74.31 ± 1.50	67.12 ± 0.97	67.25 ± 1.09
PAMF	A	85.41 ± 0.74	44.42 ± 7.22	85.37 ± 0.76	77.13 ± 0.85	76.76 ± 0.95
PAMF	C	84.86 ± 0.80	71.96 ± 0.44	84.91 ± 0.74	76.48 ± 0.45	76.18 ± 0.52
PAMF	Native T1	85.48 ± 0.87	71.53 ± 1.02	85.58 ± 0.92	78.21 ± 0.84	78.12 ± 0.85

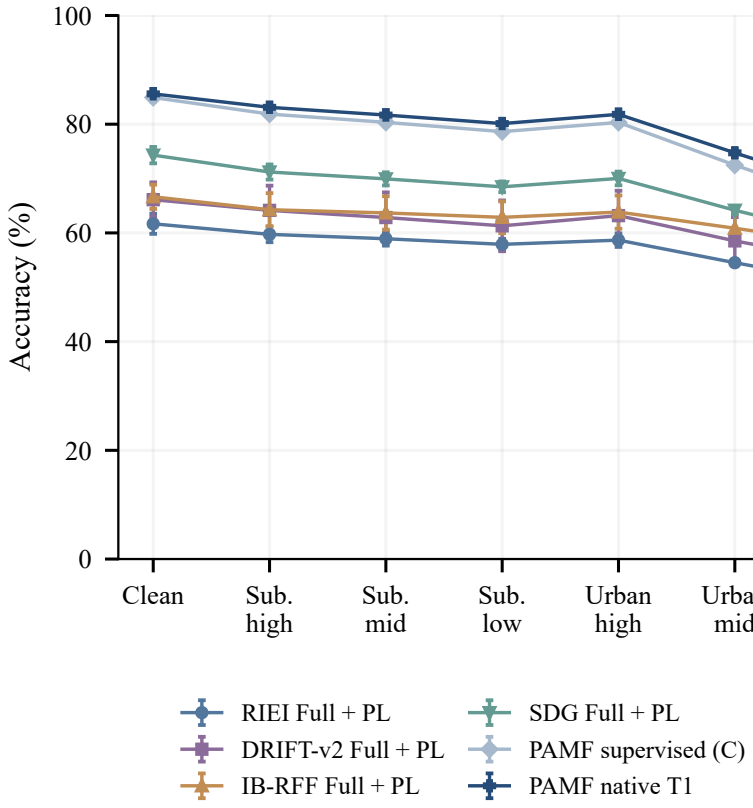


Fig. 1. F0 cross-RX accuracy on every frozen observation. External D uses the common residual-channel/ordinary-PL recipe; PAMF-Sup C uses its supervised residual-channel recipe and native T1 retains its fixed training method. This plot shows the matching DURA route; the accompanying independent direct-F figure is included in the reproducible figure kit. Error bars are three-F-seed SD.

removal per group. Each removal retains the supervised CE dose and update schedule; C-off performs no U loading or EMA work.

B increases raw clean accuracy by 1.25 ± 1.09 pp and restored S6 by 1.01 ± 0.80 pp in the complete combination. A and C have negative mean conditional restored-S6 differences (-0.41 ± 0.38 and -0.21 ± 0.58 pp for full minus removal).

TABLE IV
PAMF GENERALIZATION AFTER MATCHING G (S6 ACCURACY, %). SUPERVISED A/C USE G-L; NATIVE T1 USES G-LU. EACH ENTRY IS THE THREE-F-SEED MEAN±SD.

Split	PAMF recipe	Cross date	Cross RX	Cross date + RX
F0	Supervised, channel off	90.44 ± 0.26	77.13 ± 0.85	75.72 ± 0.11
F0	Supervised, channel on	91.20 ± 0.16	76.48 ± 0.45	74.49 ± 0.97
F0	Native T1	91.93 ± 0.07	78.21 ± 0.84	75.95 ± 0.88
F1	Supervised, channel off	83.43 ± 0.59	66.05 ± 0.92	66.31 ± 0.86
F1	Supervised, channel on	83.04 ± 0.92	65.87 ± 0.50	65.47 ± 0.22
F1	Native T1	84.48 ± 0.95	66.96 ± 0.43	65.84 ± 0.11
F2	Supervised, channel off	86.45 ± 0.08	75.16 ± 1.01	75.00 ± 0.84
F2	Supervised, channel on	87.23 ± 0.49	74.22 ± 0.78	74.65 ± 1.14
F2	Native T1	88.52 ± 0.76	74.53 ± 0.15	74.49 ± 0.26

TABLE V
CROSS-RX S6 ACCURACY (%) ACROSS DATASET/LABEL CONDITIONS. COLUMNS GIVE EACH CONDITION SEPARATELY; CLASSES ARE SIX FOR MANYSIG AND TEN FOR MANYRX.

	F0, 10% L		F0, 1% L		F0, 5% L		ManyRx	
Method	Raw	G+F	Raw	G+F	Raw	G+F	Raw	G+F
RIEI D	49.95 ± 1.55	56.73 ± 0.91	26.20 ± 8.61	33.99 ± 18.93	42.90 ± 4.07	41.11 ± 1.49	25.58 ± 1.98	28.70 ± 5.43
DRIFT D	56.76 ± 2.29	60.75 ± 4.42	17.39 ± 0.62	22.33 ± 2.03	35.73 ± 1.15	40.26 ± 8.18	34.09 ± 3.23	43.34 ± 3.64
IB-RFF D	52.57 ± 2.17	62.21 ± 2.86	45.27 ± 3.63	64.34 ± 2.47	51.83 ± 0.65	61.69 ± 0.97	48.59 ± 2.74	64.99 ± 0.77
SDG D	65.30 ± 1.14	67.12 ± 0.97	62.34 ± 0.08	68.31 ± 1.09	63.62 ± 1.48	67.06 ± 0.94	61.40 ± 0.54	66.07 ± 0.75
PAMF-Sup C	71.96 ± 0.44	76.48 ± 0.45	68.83 ± 1.10	69.26 ± 0.21	70.98 ± 0.54	74.51 ± 0.47	64.56 ± 0.13	71.40 ± 0.50
PAMF-T1	71.53 ± 1.02	78.21 ± 0.84	70.59 ± 0.12	74.04 ± 0.73	71.33 ± 0.49	75.99 ± 0.69	63.54 ± 0.39	72.59 ± 0.22

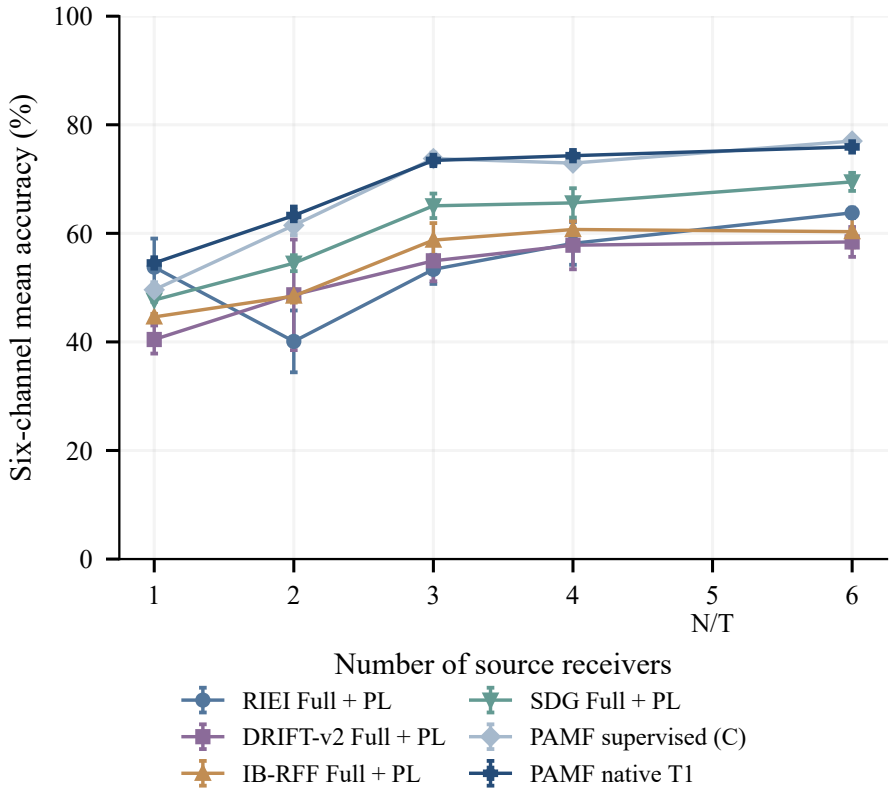


Fig. 2. Matching-DURA cross-RX natural-budget source-count curves, with fixed six-target populations and the same preregistered recipes as Table V. Scenes average within seed before the three-seed SD. Decreases and method-rank changes are retained.

The data therefore do not support a claim that all three groups yield stable positive gains in this fixed F0 setting. These results retain the designated default and are not used to regroup objectives or select reruns. The complete eight-combination

TABLE VI
PAMF CORE-GROUP ABLATION ON F0 (ACCURACY, %). ALL RESTORED ROWS USE FIXED G-LU; C-OFF REMOVES U FROM F ONLY.

F training setting	Raw clean	Raw S6	DURA clean	DURA S6
Full PAMF-T1	84.39 ± 0.92	70.60 ± 1.06	84.48 ± 0.97	77.65 ± 0.85
Without A: CRIC	84.73 ± 0.51	70.99 ± 1.24	84.78 ± 0.58	78.06 ± 0.48
Without B: consistency	83.14 ± 0.76	70.50 ± 0.61	83.13 ± 0.82	76.63 ± 0.48
Without C: selective SSL	84.79 ± 0.82	70.60 ± 0.95	84.86 ± 0.87	77.85 ± 0.40

diagnostic is retained in the experiment archive rather than expanded into the main comparison.

2) Physical Structure and Inductive Bias: Table VII isolates architecture under a common labeled-only objective. The single SIG_TIME variant changes the number of learned views; branch deletions change capacity and computation. No-time, covariance, shuffled-time and scrambled-reference controls retain parameter count but change representation or input structure. They distinguish useful response/statistical features from a claim that a specific coordinate ordering is indispensable.

Removing PA, signature or reference-response features reduces G-L S6 accuracy relative to the full backbone by 2.79 ± 0.86, 2.17 ± 0.11 and 1.48 ± 0.01 pp. However, shuffled time exceeds the original ordering after restoration, and covariance X3 is slightly higher on average. These counterexamples are retained: physical structure supplies an inductive bias, not a guarantee that every chosen coordinate or prior is optimal.

The frozen source diagnostics use the same 720 labeled packets from 90 TX/RX/day strata and 11 fixed interventions. They access neither U nor target data and fit no probe. For full T1, same-TX cross-RX feature distance is 0.0618 ± 0.0105, compared with 0.0685 ± 0.0118 without A, while the separation margin does not increase. Lower within-identity distance alone does not establish complete receiver/channel disentanglement. Simulated PA interventions test sensitivity and are not assumed to preserve unknown hardware identity.

3) Restoration Mechanisms: Table VIII retains the coarse restoration, conjugate-gradient solver, conjugate-image action and residual-input controls that directly describe DURA. All use frozen T1 default heads, identical seven-view observations and source-only G training. Coarse restoration alone is insufficient; removing CG, image action or residual input also reduces the restored S6 mean. Initialization and distillation diagnostics remain in the archive and do not occupy a separate external-comparator analysis.

D. Computational Efficiency

All inference measurements use frozen seed01 weights, FP32 with TF32 disabled, two CPU threads and exclusive RTX3090 access. Five measurement rounds describe timing variation, not five model seeds. Table IX includes complete external identifiers F, the proposed identifier and the restoration components. The external F-only measurements do not establish their online G+F latency; cached classification time is not substituted for it. External timing rows are historical architecture/execution references measured before the practical-residual retraining, not new checkpoint-specific latency measurements. The default PAMF F+G route has GPU-wall median 4.541ms and P95 4.725ms; CPU class return has median 4.611ms. The same weights take 37.190ms on the native execution path, giving an $8.189\times$ execution-route improvement. This is not a claim that every external model received identical implementation optimization.

F and G contain 232993 and 266951 registered learned parameters (499944 total). Separate eager Conv/Linear profiles give 8.856M and 25.068M real MACs; these omit functional-head arithmetic, FFT complex operations, CG reductions and elementwise work, so they are not total FLOPs. Eager G uses 43 FFT/IFFT calls; the residual-reuse route uses 27. The larger training wrapper, optimizer and EMA are not counted as deployment parameters. Multi-model/Graph resident allocation is not reported as single-model VRAM.

Table IX(b) preserves the nonmonotonic batch latency. Amortized throughput does not establish continuous full-rate radio operation. Training-only CRIC, stochastic forwards, EMA and U gates add no deployment passes. Source-stage cost, G ancestry, cached inference and scoring costs are reported separately in the evidence archive.

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TABLE VII
PHYSICAL-STRUCTURE ABLATION UNDER MATCHED LABELED-ONLY CE PLUS SATELLITE AUGMENTATION: NO CRIC/KL/EMA/U, 11100 UPDATES (ACCURACY, %). THE FULL PAMF-SUP RECIPE IS EVALUATED SEPARATELY.

PAMF backbone	Raw clean	Raw S6	G-L S6	G-L S6 F1
Full X3 backbone	81.55 ± 1.34	46.35 ± 1.74	74.49 ± 1.14	73.91 ± 1.29
Single SIG_TIME	81.82 ± 1.26	43.61 ± 6.18	74.20 ± 1.16	73.72 ± 1.33
Without time coordinate	80.95 ± 1.54	45.46 ± 5.89	73.99 ± 1.03	73.39 ± 1.27
Without signature branch	79.48 ± 1.72	41.84 ± 5.75	72.32 ± 1.09	71.92 ± 1.09
Without PA branch	78.94 ± 2.29	39.39 ± 5.78	71.70 ± 0.44	71.13 ± 0.49
Without reference response	78.64 ± 1.49	44.80 ± 2.93	73.01 ± 1.15	72.32 ± 1.16
Covariance X3	81.69 ± 0.61	49.36 ± 4.72	74.56 ± 0.84	73.89 ± 0.89
Shuffled time	82.20 ± 1.72	48.54 ± 6.44	75.62 ± 1.11	75.10 ± 1.24
Scrambled reference	81.02 ± 1.20	46.59 ± 2.91	74.24 ± 1.09	73.64 ± 1.24

TABLE VIII
DURA FUNCTIONAL ABLATION (ACCURACY, %). DEFAULT: THREE F SEEDS WITH ONE G; OTHER ROWS: SAME-INDEX F/G STUDENTS. SD DESCRIBES THE RESPECTIVE SEED DESIGN.

Restorer	Parameters (K)	Clean	S6	Low urban
Default DURA	267.0	84.48 ± 0.97	77.65 ± 0.85	67.66 ± 0.71
Coarse only	106.1	83.26 ± 1.02	72.11 ± 0.77	55.95 ± 1.24
Without CG	267.0	84.41 ± 1.08	76.94 ± 1.08	66.45 ± 1.31
Without image action	267.0	84.49 ± 0.96	77.32 ± 1.10	67.09 ± 1.38
Without residual input	267.0	84.38 ± 0.94	76.78 ± 1.21	66.65 ± 1.53

TABLE IX
INFERENCE ON RTX3090 (MS, FIVE ROUNDS). (A) BATCH-ONE GPU WALL AND CPU-RETURN LATENCY; EXTERNAL F IS EAGER, ONLINE EXTERNAL G+F IS UNMEASURED. (B) PAMF F+G THROUGHPUT IN 256-SAMPLE IQ WINDOWS/S.

(a) Batch-one latency

Model	Execution	Median	P95	CPU return
RIEI F	Eager	2.352	2.734	2.456
DRIFT-v2 F	Eager	2.408	2.605	2.476
IB-RFF F	Eager	1.456	2.168	1.506
SDG F	Eager	3.054	5.008	3.105
PAMF-T1 F	CUDA Graph	0.757	0.858	0.796
DURA G	Graph + residual reuse	3.843	3.984	3.874
PAMF-T1 F+G	Graph + residual reuse	4.541	4.725	4.611

(b) Batch throughput

Batch	Median	IQ/s
1	4.541	220.2
8	13.747	581.9
32	28.374	1127.8
64	15.295	4184.5
128	17.111	7480.5